

ITEC 3100 – Introduction to Networks

Professor: Dr. Binh Tran

Lab 5:

Please use a different color font for your answers so I can clearly see them:

Q1. What is your definition of a network protocol and why are they needed?

Rules and procedures and they are needed for communication

Q2. How many layers exist in the OSI model and how many in the TCP/IP model? Which one is a more practical model?

Osi-7

Tcp/ip-4

Tcp/ip is more practical

Q3. Which layer of both models do the protocols IP, ICMP, ARP and IPSec operate at?

Osi-layer 3

Tcp/ip-layer 2

Q4. Which layer of both models do the protocols TCP and UDP operate at?

Osi-layer 4

Tcp-layer3

Q5. What is the difference between TCP and UDP? Provide a real world example?

Udp is connectionless and tcp is connected, a real world example of udp is playing an online game and an example of tcp is sending a email

Watch this DNS animated video:

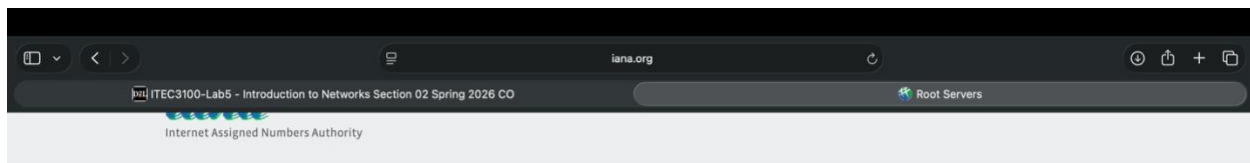
<https://www.youtube.com/watch?v=mpQZVYPuDGU>

Q6. Explain in your own words how DNS operates and what is referred to as the “root”?

DNS is the in between of how we interact with websites and how computers interact with websites. The Root Servers are the top level of the DNS hierarchy

Q7. How many root servers exist and what are their names? (Provide reference link). 13

<https://www.iana.org/domains/root/servers>



Root Servers

The authoritative name servers that serve the DNS root zone, commonly known as the "root servers", are a network of hundreds of servers in many countries around the world. They are configured in the DNS root zone as 13 named authorities, as follows.

List of Root Servers

HOSTNAME	IP ADDRESSES	OPERATOR
a.root-servers.net	198.41.0.4, 2001:503:ba3e::2:30	Verisign, Inc.
b.root-servers.net	170.247.170.2, 2801:1b8:10::b	University of Southern California, Information Sciences Institute
c.root-servers.net	192.33.4.12, 2001:500:2::c	Cogent Communications
d.root-servers.net	199.7.91.13, 2001:500:2d::d	University of Maryland
e.root-servers.net	192.203.230.10, 2001:500:a8::e	NASA (Ames Research Center)
f.root-servers.net	192.5.5.241, 2001:500:2f::f	Internet Systems Consortium, Inc.
g.root-servers.net	192.112.36.4, 2001:500:12::d0d	US Department of Defense (NIC)
h.root-servers.net	198.97.190.53, 2001:500:1::53	US Army (Research Lab)
i.root-servers.net	192.36.148.17, 2001:7fe::53	Netnod
j.root-servers.net	192.58.128.30, 2001:503:c27::2:30	Verisign, Inc.
k.root-servers.net	193.0.14.129, 2001:7fd::1	RIPE NCC
l.root-servers.net	199.7.83.42, 2001:500:9f::42	ICANN
m.root-servers.net	202.12.27.33, 2001:dc3::35	WIDE Project

Configuring the Root Servers

Watch this DHCP Animated video:

<https://www.youtube.com/watch?v=e6-TaH5bkjo>

Q8. Explain in your own words how DHCP operates:

DHCP gives an ip to a device that joins a network

Q9. What is the difference between static versus dynamic IP addresses? When should we use each?

Static is manual

Dynamic is automatic

We should use dynamic

Q10. Run the **ipconfig /all** command and insert screenshot:

```
ether 36:aa:f5:1f:0f:04
media: autoselect <full-duplex>
status: inactive
bridge0: flags=8863<UP, BROADCAST, SMART, RUNNING, SIMPLEX, MULTICAST> mtu 1500
options=63<RXCSUM, TXCSUM, TS04, TS06>
ether 36:aa:f5:1f:0f:00
Configuration:
    id 0:0:0:0:0:0 priority 0 hellotime 0 fwddelay 0
    maxage 0 holdcnt 0 proto stp maxaddr 100 timeout 1200
    root id 0:0:0:0:0:0 priority 0 ifcost 0 port 0
    ipfilter disabled flags 0x0
    member: en1 flags=3<LEARNING, DISCOVER>
        ifmaxaddr 0 port 0 priority 0 path cost 0
    member: en2 flags=3<LEARNING, DISCOVER>
        ifmaxaddr 0 port 0 priority 0 path cost 0
    nd6 options=201<PERFORMNUD, DAD>
    media: <unknown type>
status: inactive
api1: flags=8822<BROADCAST, SMART, SIMPLEX, MULTICAST> mtu 1500
options=400<CHANNEL_IO>
ether 32:f5:d4:9d:b4:cd
nd6 options=201<PERFORMNUD, DAD>
media: autoselect (none)
en0: flags=8863<UP, BROADCAST, SMART, RUNNING, SIMPLEX, MULTICAST> mtu 1500
options=6406<TS04, TS06, CHANNEL_IO, PARTIAL_CSUM, ZEROINVERT_CSUM>
ether 08:00:a8:3f:2:a9
inet6 fe80::1c4:d87:c83:60f2%en0 prefixlen 64 secured scopeid 0xb
inet6 fd5f:8937:d84:5827:1015:fd7:7326:475c prefixlen 64 autoconf secured
inet 192.168.86.40 netmask 0xfffff000 broadcast 192.168.86.255
nd6 options=201<PERFORMNUD, DAD>
media: autoselect
status: active
awd10: flags=8863<UP, BROADCAST, SMART, RUNNING, SIMPLEX, MULTICAST> mtu 1500
options=6406<TS04, TS06, CHANNEL_IO, PARTIAL_CSUM, ZEROINVERT_CSUM>
ether b6:d4:88:6c:8d:a8
inet6 fe80::1b4d:a8ff:fe6c:8da8%awd10 prefixlen 64 scopeid 0xd
nd6 options=201<PERFORMNUD, DAD>
media: autoselect
status: active
llw0: flags=8863<UP, BROADCAST, SMART, RUNNING, SIMPLEX, MULTICAST> mtu 1500
options=400<CHANNEL_IO>
ether b6:d4:88:6c:8d:a8
inet6 fe80::1b4d:a8ff:fe6c:8da8%llw0 prefixlen 64 scopeid 0xe
nd6 options=201<PERFORMNUD, DAD>
media: autoselect (none)
utun0: flags=8851<UP, POINTOPOINT, RUNNING, MULTICAST> mtu 1500
inet6 fe80::a914:f981:aace:4b0b%utun0 prefixlen 64 scopeid 0xf
nd6 options=201<PERFORMNUD, DAD>
utun1: flags=8851<UP, POINTOPOINT, RUNNING, MULTICAST> mtu 1380
inet6 fe80::d6c:993d:c52:b349%utun1 prefixlen 64 scopeid 0x10
nd6 options=201<PERFORMNUD, DAD>
utun2: flags=8851<UP, POINTOPOINT, RUNNING, MULTICAST> mtu 2000
inet6 fe80::cbe0:4f53:66ad:68a8%utun2 prefixlen 64 scopeid 0x11
nd6 options=201<PERFORMNUD, DAD>
utun3: flags=8851<UP, POINTOPOINT, RUNNING, MULTICAST> mtu 1800
inet6 fe80::ce81:b1c:bd2c:69e%utun3 prefixlen 64 scopeid 0x12
nd6 options=201<PERFORMNUD, DAD>
utun4: flags=8851<UP, POINTOPOINT, RUNNING, MULTICAST> mtu 1380
inet6 fe80::cc7b:d8cd:fad:b28%utun4 prefixlen 64 scopeid 0x13
nd6 options=201<PERFORMNUD, DAD>
utun5: flags=8851<UP, POINTOPOINT, RUNNING, MULTICAST> mtu 1380
inet6 fe80::92d1:258d:ffae:d5d1%utun5 prefixlen 64 scopeid 0x16
nd6 options=201<PERFORMNUD, DAD>
shill@mac ~ %
```

Q11. What does ARP do? Run the **arp -a** command and insert screenshot: **bridge between layer 3 and layer 2 (osi)**

```
[shill@mac ~ % arp -a
? (192.168.86.1) at 24:29:34:d4:f1:23 on en0 ifscope [ethernet]
? (192.168.86.27) at 3a:ef:7a:2b:dd:f7 on en0 ifscope [ethernet]
iphone.lan (192.168.86.28) at 9e:d8:6d:c9:90:de on en0 ifscope [ethernet]
takiyahs-air.lan (192.168.86.29) at 14:7f:ce:90:8b:53 on en0 ifscope [ethernet]
iphone.lan (192.168.86.30) at 4e:65:11:9b:65:5a on en0 ifscope [ethernet]
mac.lan (192.168.86.40) at ee:d9:a8:a3:f2:a9 on en0 ifscope permanent [ethernet]
? (192.168.86.255) at ff:ff:ff:ff:ff:ff on en0 ifscope [ethernet]
mdns.mcast.net (224.0.0.251) at 1:0:5e:0:0:fb on en0 ifscope permanent [ethernet]
shill@mac ~ %
```

Q12. Run the **ping 8.8.8.8** command and insert screenshot:

```
shill@mac ~ % ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8): 56 data bytes
64 bytes from 8.8.8.8: icmp_seq=0 ttl=115 time=19.497 ms
64 bytes from 8.8.8.8: icmp_seq=1 ttl=115 time=20.798 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=115 time=19.954 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=115 time=15.809 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=115 time=16.076 ms
64 bytes from 8.8.8.8: icmp_seq=5 ttl=115 time=17.037 ms
64 bytes from 8.8.8.8: icmp_seq=6 ttl=115 time=14.876 ms
64 bytes from 8.8.8.8: icmp_seq=7 ttl=115 time=17.287 ms
64 bytes from 8.8.8.8: icmp_seq=8 ttl=115 time=16.499 ms
64 bytes from 8.8.8.8: icmp_seq=9 ttl=115 time=27.128 ms
64 bytes from 8.8.8.8: icmp_seq=10 ttl=115 time=15.037 ms
64 bytes from 8.8.8.8: icmp_seq=11 ttl=115 time=14.534 ms
64 bytes from 8.8.8.8: icmp_seq=12 ttl=115 time=14.082 ms
64 bytes from 8.8.8.8: icmp_seq=13 ttl=115 time=15.961 ms
64 bytes from 8.8.8.8: icmp_seq=14 ttl=115 time=15.476 ms
64 bytes from 8.8.8.8: icmp_seq=15 ttl=115 time=16.505 ms
64 bytes from 8.8.8.8: icmp_seq=16 ttl=115 time=14.203 ms
64 bytes from 8.8.8.8: icmp_seq=17 ttl=115 time=15.855 ms
64 bytes from 8.8.8.8: icmp_seq=18 ttl=115 time=14.098 ms
64 bytes from 8.8.8.8: icmp_seq=19 ttl=115 time=16.244 ms
```

Q13. Run **tracert** www.google.com and insert screenshot:

```
Last login: Tue Mar 17 09:28:09 on ttys000
shill@mac ~ % traceroute www.google.com
traceroute: Warning: www.google.com has multiple addresses; using 64.233.185.103
traceroute to www.google.com (64.233.185.103): 64 hops max, 40 byte packets
 1 192.168.86.1 (192.168.86.1)  0.322 ms  4.586 ms  0.424 ms
 2 * * *
```

Q14. What is the difference between IPv4 and IPv6 addresses?

Capacity, security, configuration

Q15. Why does the US still use IPv4 addresses while many other countries only use IPv6?

Have a lot of ipv4 equipment that will cost a lot to replace

Complete and upload to the Lab 5 dropbox.